

depends on the maximum energy of the incident electrons.

- 29** The fusion of two deuterium nuclei produces a nuclide of helium and a neutron. The reaction liberates 3.27 MeV of energy.

How does the combined mass of the two deuterium nuclei, $\Sigma M_{\text{Reactants}}$, compare with the combined mass of the helium nucleus and neutron, $\Sigma M_{\text{Products}}$?

- A** $\Sigma M_{\text{Reactants}}$ is 5.8×10^{-30} kg greater than $\Sigma M_{\text{Products}}$.
- B** $\Sigma M_{\text{Reactants}}$ is 5.8×10^{-30} kg smaller than $\Sigma M_{\text{Products}}$.
- C** $\Sigma M_{\text{Reactants}}$ is 5.8×10^{-36} kg greater than $\Sigma M_{\text{Products}}$.
- D** $\Sigma M_{\text{Reactants}}$ is 5.8×10^{-36} kg smaller than $\Sigma M_{\text{Products}}$.

Ans: A

$$3.27 \text{ MeV} = 5.23 \times 10^{-30} \text{ J}$$

$$\text{Mass equivalent} = 5.8 \times 10^{-30} \text{ kg}$$

Greater mass before fusion to liberate the energy.

30 The graph shows how the binding energy per nucleon of a nucleon of a nucleus varies with